Solution

Matrix Theory, Quiz 6K — March 27, 2026

Given the matrix A and vector $\mathbf{v}$

$$A = \begin{bmatrix} 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 1 & -1 & -1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -2 \\ 2 \\ -5 \\ 1 \end{bmatrix}$$

answer the following.

(1) Find a basis for the column space of A.
(2) Find a basis for the row space of A.
(3) Find a basis for the null space of A.
(4) Determine whether the vector $\mathbf{v}$ is in the column space of A. If yes, write $\mathbf{v}$ as a linear combination of the basis given in (1).
(5) Extend the basis given in (1) to a basis for $\mathbb{R}^4$.

4

reduce $(A|v)$ to RREF

$$[A|v] = \left[\begin{array}{cccc|c} 0 & -1 & 0 & -1 & -2 \\ 0 & 0 & 1 & -1 & 2 \\ 1 & -1 & -1 & -1 & -5 \\ -1 & 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow[R_3+R_4]{-R_4,\ -R_1,\ R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & -1 & -1 & 0 & -4 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & 1 & -2 \end{array}\right]$$

$$\to \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

1 (1) Rank $A = 3$ basis col sp $A = c_1, c_2, c_3$ of A

1 (2) basis row sp. A $= \{(1,0,0,-1), (0,1,0,1), (0,0,1,-1)\}$

(3) Null $A = 1$

1 Null sp $A = d\begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}$ basis Null sp $\left\{\begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}\right\}$

(4) Yes $A\bar{x} = \bar{v}$ is consistent. $\bar{x} = \begin{pmatrix} -1 \\ 2 \\ 2 \\ 0 \end{pmatrix}$

2 $v = -\begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} + 2\begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$

(5) adjoin e_1 to c_1, c_2, c_3

1

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

rank $= 4$

Solution

Matrix Theory, Quiz 6L — March 27, 2026

Given the matrix A and vector $\mathbf{v}$

$$A = \begin{bmatrix} 0 & 0 & 1 & -1 \\ 1 & -1 & -1 & -1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ -4 \\ 1 \end{bmatrix}$$

answer the following.

(1) Find a basis for the column space of A.
(2) Find a basis for the row space of A.
(3) Find a basis for the null space of A.
(4) Determine whether the vector $\mathbf{v}$ is in the column space of A. If yes, write $\mathbf{v}$ as a linear combination of the basis given in (1).
(5) Extend the basis given in (1) to a basis for $\mathbb{R}^4$.

$$(A|v) = \left[\begin{array}{cccc|c} 0 & 0 & 1 & -1 & 1 \\ 1 & -1 & -1 & -1 & 4 \\ -1 & 0 & 0 & 1 & -4 \\ 0 & -1 & 0 & -1 & 1 \end{array}\right] \xrightarrow[R_1 \quad -R_4 \quad R_2+R_3]{-R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 4 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & -1 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 4 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & -1 & 1 & -1 \end{array}\right]$$

4

$$\to \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 4 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

(1) Rank $A = 3$ basis col sp $A = c_1, c_2, c_3$ of A

1

(2) basis Row sp $A = R_1, R_2, R_3$ of RREF

1 $= \{(1,0,0,-1), (0,1,0,1), (0,0,1,-1)\}$

1 (3) Null $A = 1$ Null Sp. $A = \alpha \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}$ basis Null Sp $\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix} \right\}$

(4) Yes: $A\bar{x} = \bar{v}$ consistent $\bar{x} = \begin{pmatrix} 4 \\ -1 \\ 1 \\ 0 \end{pmatrix}$

2

$$\bar{v} = 4\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}$$

(5) adjoin e_1 to c_1, c_2, c_3

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \xrightarrow{R_3+R_2} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \xrightarrow{R_4-R_3} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1

rank $= 4$